Supplementary Appendix for "Top-k Identification from Noisy Pairwise Comparisons under Restricted Comparison Graphs"

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1 Implementation details of Algorithm 1 and experimental parameters

Algorithm 1 (GraphWalk-SE, Section 5 of the main text) uses calibrated devices that are not part of Theorem 5.1: active-set query concentration, and contrast radii $\tau_{ij} = \sqrt{2\log(4n^2r/\delta)\, s_{ij}^2} + 2\log(4n^2r/\delta)\, s_{ij}^2$ with $s_{ij}^2 = P_{ii} + P_{jj} - 2P_{ij}$, $P = H^+$ the pseudo-inverse of the observed Fisher information, per-edge curvature $\sigma'(\hat{d}_e)$ clipped to $|\hat{d}_e| \le 3$, a round budget that doubles, and 10% full-graph mixing in opponent sampling. A capped run (E4) finishes its current doubling round, so realized T can exceed the cap by up to one round. All error rates are measured on fresh random seeds.

Experimental parameters and reproducibility. Table 1 lists the parameters of every experiment. Instances are drawn once per configuration (not per repetition), as $\theta_i = 3\tanh\big((k - \frac{1}{2} - i)\,\Delta/3\big)$; repetitions differ only through algorithmic randomness, seeded deterministically from (experiment, configuration, repetition index), so every run is reproducible from the code in the accompanying repository (which also archives all `results/*.json` files, Python and NumPy versions). Subsections of the main text are ordered by narrative rather than by experiment number.

Table 1: Parameters of all experiments. "Alg." is A for Algorithm 1 (GraphWalk-SE) and U for UE-topk.

exp	graph(s)	n	k	Δ grid (points)	reps, Alg
E1	complete	10, 20, 50	3	0.2, 0.5, 1.0 (3)	100, A
E2	complete, random (p=0.15), star, dumbbell, cycle	50	3	0.5 (1)	100, A
E2b	complete, random (p=0.2), star, dumbbell, cycle	20	3	0.5 (1)	50, U
E2c	complete, dumbbell	50	25	0.5 (1)	50, U
E2d	complete	10, 20, 50	3	0.5 (1)	50, U
E3	complete	20	3	0.1 to 1.5 (15)	100, A
E3-fine	complete	20	3	0.05 to 2.0 (24, log-uniform)	200, A
E3-UE	complete	20	3	0.05 to 2.0 (8, log-uniform)	50, U
E4	complete	20	3	0.5 (1), caps 1000 to 16000	100, A
E5	complete, cycle	20	3	0.5 (1), budgets 500 to 128000	100, both